Roll No.

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharma (Sem-3)

PHARMACEUTICAL MICROBIOLOGY

Subject Code : BP-303T

M.Code : 75107

Date of Examination : 06-06-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Answer briefly :**

   a) What is electron microscopy?

   b) Define acid-fast staining.

   c) Define disinfectant with an example.

   d) Give a method for Indole production test in biochemical testing.

   e) Define aseptic area.

   f) What is microbial assay?

   g) What are the nutritional requirements of bacteria?

   h) Write the name and size of the smallest and largest virus?

   i) What is a membrane filter and give their types?

   j) Enlist the various sterility indicators used in sterility monitoring.

SECTION-B

2. Discuss in detail about the of gaseous and radiation method of sterilization.

3. Define disinfectant and how to evaluate it by phenol coefficient test?

4. **Explain about :**

   a) Counting techniques of bacteria.

   b) Method of isolation of bacteria.

SECTION-C

5. Diagrammatically explain the phase of contrast microscope.

6. Write methods for sterility testing of ophthalmic formulations according to IP.

7. What are the different methods of microbial assay of antibiotics?

8. Difference between dry heat and moist heat sterilization.

9. Write down the types of microbial spoilage with examples.

10. What are the methods for the preservation of pure culture?

11. What is growth curve of bacteria?

12. Write a method for gram staining for the identification of microbes.

13. Applications of cell culture in pharmaceutical industry.

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NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

July-2023